Introduction

We hope to solve the equation

$$i\partial_t \psi = (-i\sigma_x \partial_x + m\sigma_z + f(x))\psi$$

where $\psi = (\psi_1 \psi_2)^T$, and f is some potential function. In theory, this is only slightly different from Schrödinger. If we can store the spinor ψ, nearly all the same methods can be used to solve Dirac.

Discretization and Storage

Fortunately, the problem of storing the spinor on qubits has a simple solution: we maintain a register to store position, as in Schrödinger, and add one qubit as an indicator. We discretize ψ_1 and ψ_2 into $|\psi_1\rangle$ and $|\psi_2\rangle$ as we know how from Schrödinger, and store the state ψ as

$$\begin{aligned}
|\psi\rangle &\propto \sum_{k=0}^{N-1} (\psi_1(x_k)\,|0\rangle + \psi_2(x_k)\,|1\rangle) \otimes |k\rangle \\
&= |0\rangle \otimes \left(\sum_{k=0}^{N-1} \psi_1(x_k)\,|k\rangle \right) + |1\rangle \otimes \left(\sum_{k=0}^{N-1} \psi_2(x_k)\,|k\rangle \right) \\
&\propto |0\rangle \otimes |\psi_1\rangle + |1\rangle \otimes |\psi_2\rangle
\end{aligned}$$

With the indicator as the most significant qubit, this is equivalent to concatenating the column vectors of amplitudes for $|\psi_1\rangle$ and $|\psi_2\rangle$.